

Divanshu Sharma

Machine Learning Engineer

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SKILLS

Languages:	Python, C++, Bash
Libraries & Frameworks:	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, PyTorch Geometric (PyG), SciPy, OpenCV, BeautifulSoup
MLOps & Deployment:	Docker, CI/CD (GitHub Actions), MLflow, Airflow, FastAPI, REST API, Streamlit, Gradio
Cloud Platforms:	AWS (SageMaker, EC2), GCP (AI Platform), Azure (basic)
Data:	PostgreSQL, MySQL, MongoDB

EXPERIENCE

Research Consultant | WorldQuant BRAIN | Nov 2022 – Present

- Developed, backtested, and optimized over 130 alpha models for global equity markets, directly contributing to an 8 - 10% increase in the portfolio's Sharpe ratio through an advanced ensemble "Super Alpha" approach.
- Engineered robust validation pipelines for models using over 10 years of tick-level financial data, applying walk-forward validation and regularization techniques to significantly reduce overfitting.

PROJECTS & OPEN-SOURCE

ML4SCI Task Solutions Repository (CMS) | [GitHub](#)

Technologies: PyTorch, PyTorch Geometric, TensorFlow, CNN, GCN

- Designed and created a custom ResNet-15 CNN architecture that achieved over 90% AUC in classifying electrons versus photons from raw particle detector data, contributing to more accurate event reconstruction in high-energy physics simulations.
- Implemented a Graph Convolutional Network (GCN) for particle momentum regression, which successfully reduced model training time by 30% while preserving equivalent ROC performance compared to a more complex GAT baseline.
- Enhanced quark/gluon jet classification accuracy by 7% by developing custom VGG-style CNNs and systematically applying targeted regularization techniques to mitigate overfitting on sparse, high-dimensional data.

CNN-based Accident Detection Web-App | [GitHub](#)

Technologies: YOLOv8, PyTorch, TensorFlow, Flask, Docker, Roboflow

- Spearheaded and deployed a state-of-the-art YOLOv8 model for real-time accident detection, achieving a high mean Average Precision (mAP) of 83% and a rapid 50ms per-frame inference speed suitable for live video streams.
- Built and containerized a Flask-based REST API to serve the learned model, integrating a Roboflow-exported dataset of over 1,200 annotated images to achieve a 0.87 F1-score.

Next Word Prediction | [Kaggle](#)

Technologies: TensorFlow, Keras, RNN, LSTM

- Engineered a multi-layer LSTM network leveraging a 580K-sentence corpus to learn and predict subsequent words, achieving 92% next-token accuracy and decreasing model perplexity by 35%.

EDUCATION

University of Mumbai | Mumbai, India

Bachelor of Engineering, Computer Science | Expected June 2025 | CGPA: 8.2/10.0

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Probability & Statistics

Qiskit Global Summer School | July 2022

- Completed an intensive, two-week program focused on Quantum Simulation and algorithms, earning a final grade of Quantum Excellence for outstanding project work and comprehension.

ACHIEVEMENTS & CERTIFICATIONS

- Machine Learning Specialization & Deep Learning Specialization (DeepLearning.AI, 2023)
- TensorFlow Developer Professional Certificate (DeepLearning.AI, 2023)
- Generative Adversarial Networks (GANs) Specialization (DeepLearning.AI, 2024)
- GOLD LEVEL** (Top 1 %), **STAGE 1 Qualifier**, WorldQuant BRAIN Global Alphathon 2022: Achieved a top-tier ranking among thousands of global participants by designing and submitting high-performing quantitative trading models.
- Kaggle 2x Expert